

Multiple Choice (10 marks)

1. Consider $x(t)$ to be given as, $x(t) = 10 \cos(20\pi - \pi/4) - 5 \cos(50\pi t)$ What is minimum sampling rate (/H)?
 - (a) 30
 - (b) 20
 - (c) 40
 - (d) 50
 - (e) 35
2. Which types of functions grow the slowest?
 - (a) $O(N)$
 - (b) $O(\log \log N)$
 - (c) $O(N!)$
 - (d) $O(\log N)$
 - (e) $O(N^N)$
3. A 3-way, set-associative cache is
 - (a) only effective when more than 3 processes are running alternately on the processor
 - (b) possible only with write-through
 - (c) one in which each main memory word can be stored at any of 4 cache locations
 - (d) possible only with write-back
 - (e) effective only if 3 or fewer processes are running alternately on the processor
4. Let $x = 8$. What is $x >> 1$ in Python 3?
 - (a) 5
 - (b) 3
 - (c) 0
 - (d) 8
 - (e) 4
5. In Python 3, $b = [11, 13, 15, 17, 19, 21]$; $\text{print}(b[::2])$ outputs what?

13,17,21

13,15,17

11,15
11,13,15,19
11,15,19

True / False (10 marks)

Answer True or False

6. True or False: An excessively large step size can cause the training loss to increase with the number of epochs due to instability in learning.
7. True or False: In a simple planar graph with 30 vertices, the maximum number of edges is 84, according to Euler's formula.
8. True or False: A convex polygon with 26 faces and 39 edges has 15 vertices according to Euler's formula for polyhedra.
9. True or False: The TCP connect scan is designed to complete the TCP handshake for port scanning.
10. True or False: The use of weak classifiers in bagging helps prevent overfitting by averaging their predictions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find the first repeated character in a given string.
12. Write a function to merge multiple sorted inputs into a single sorted iterator using heap queue algorithm.

End of Paper